Whole-dissertation replication of Schwabe's psychometric approach to behaviour genetics

Abstract

We partly replicated the headline findings of Schwabe (2016), who reported an IRT remedy for scale-induced interactions and a full-information power advantage: all 108 reconstructed variance-SE contrasts favor full information after multiplicity adjustment, but all 216 literal source SE values fall outside conditional comparison bands. Unknown source Monte Carlo error and SE implementation prevent a definitive numerical replication verdict; precision gains do not establish the reported power advantage. The joint Bayesian convergence gate failed, and full recovery/power grids and independent empirical effects are not established. Eleven full-information point-bias signs reverse relative to the source; none excludes zero after adjustment. For illustration, the published deletion-minus-full A variance-SE difference is 0.00106356 versus our 0.00150235, with unresolved magnitude comparability; MAR deletion heritability bias is 0.03315402 versus 0.03118039, both positive. Conditional mathematical results support restricted model properties. Coverage of all 996 linked records is complete; whole-dissertation replication is not.

Scientific estimand / claim	Published	Ours	Effect magnitude	Direction	Verdict
HE01 No scalar: subset-invariance law	not established	not established	not established	not established	partial
HE02 No scalar: IRT recovery family	not established	not established	not established	not established	not established
HE03 No scalar: power design grid	not established	not established	not established	not established	not established
HE04 No scalar: sum-score bias family	not established	not established	not established	not established	not established
HE05 Mathematics AxE slope	2.0837	not established	not established	not established	blocked
HE06 Mathematics AxC slope	2.6159	not established	not established	not established	blocked
HE07 No scalar: two attitude dimensions	not established	not established	not established	not established	blocked
HE08 No scalar: nine-item subscale	not established	not established	not established	not established	blocked
HE09 Liberalism AxE slope	-2.81	not established	not established	not established	blocked
HE10 Liberalism AxC slope	0.54	not established	not established	not established	blocked
HE11 No scalar: AxE coefficient pair	not established	not established	not established	not established	blocked
HE12 No scalar: AxC coefficient pair	not established	not established	not established	not established	blocked
HE13 No scalar: identification theorem	not established	not established	not established	not established	partial
HE14 No scalar: measurement-bias remedy	not established	not established	not established	not established	not established
HE15 No scalar: likelihood uniqueness	not established	not established	not established	not established	not established
HE16 No scalar: A-moderation power	not established	not established	not established	not established	not established
HE17 No scalar: C-moderation power	not established	not established	not established	not established	not established
HE18 No scalar: twenty-item sufficiency	not established	not established	not established	not established	not established
HE19 High-SES shared-variance slope	-3.0777	not established	not established	not established	blocked
HE20 No scalar: 13-covariate family	not established	not established	not established	not established	blocked
HE21 No scalar: power across designs	not established	not established	not established	not established	not established
HE22 No scalar: 108 precision contrasts	not established	not established	not established	not established	partial
HE23 No scalar: observed-likelihood law	not established	not established	not established	not established	partial
HE24 No scalar: 24 MAR A bias contrasts	not established	not established	not established	not established	partial
HE25 No scalar: 24 MAR C bias contrasts	not established	not established	not established	not established	partial
HE26 No scalar: 24 MAR h2 bias contrasts	not established	not established	not established	not established	partial
HE27 Example: A SE, MCAR, q=2, 2%	0.00106356	0.00150235	not established	same direction	partial
HE28 Example: h2 bias, MAR, q=5, 10%	0.03315402	0.03118039	smaller	same direction	partial

Descriptive point estimates; not equivalence: this comparison applies only to scalar rows. "No scalar" identifies a proposition or family with no justified aggregate; its "not established" cells do not adjudicate scientific truth. The separate Verdict column distinguishes partial support, blocked refits and unresolved conclusions. Slopes: log-variance/genetic-score unit; SES: high/low-average; HE27: SE in squared phenotype units; HE28: heritability proportion. Tolerance 5e-05 for HE05/06/19; Tolerance 0.005 for HE09/10 (half a printed unit). HE27 tolerance 0.00023667095377570213; HE28 tolerance 0.00027219389842495325 (geometric bounds, not source Monte Carlo error). Sources: Schwabe (2016), Chapters 1–6; principal locators in Appendix B and all 187 evidence-only component comparisons linked in Appendix A.

1 Introduction

We partly replicated Schwabe's (2016) headline findings: all 108 reconstructed deletion-minus-full variance-SE contrasts favor full information after multiplicity adjustment, but all 216 literal source SE values fall outside conditional comparison bands. Unknown source Monte Carlo error and SE implementation prevent a definitive numerical replication verdict. The source's full-information power advantage is not established by our precision result. The joint Bayesian convergence gate failed; full Bayesian recovery and power grids and independent empirical effects are not established. Eleven full-information point-bias signs reverse relative to the source, but none excludes zero after adjustment. In the precision illustration, the published 0.00106356 becomes 0.00150235 in our reconstruction, with unresolved magnitude comparability; the MAR deletion heritability bias is 0.03315402 versus 0.03118039, both positive. Its extraction tolerance is 0.00027219389842495325. These point estimates do not establish a statistically different or equivalent effect.

The dissertation asks how measurement and missing covariates affect genetic and environmental inference. Its item response theory (IRT) models separate latent traits from observed item responses; its biometric models separate additive genetic variance (A), shared environment (C), and unique environment (E). It reports scale-induced interactions under sum scores, remedies using item-level models, empirical moderation of mathematics and liberalism, a logvariance moderation model, and gains from full-information treatment of incomplete covariates (Schwabe, 2016, Chapters 1–7). These are distinct scientific targets. Success on covariance algebra or a reconstructed simulation does not supply missing empirical observations or establish source-count Bayesian recovery.

Our coverage comprises all seven chapters and Appendices A–G, with 996 linked records and 26 principal claims. Records include parents, shared components, source-only calculations and blocked targets. They are not 996 independent experiments, and no replication success fraction is defined. The first-page table preserves all principal claims and two illustrations; its nonscalar rows represent conjunctions or propositions for which no scientific scalar aggregate exists.

2 What we did ourselves, and what we took from others

Our contribution is a condition-specific reconstruction, mathematical checks, and a source-to-result comparison. Schwabe supplied the models, published designs, tables, figures and original programs. We derived conditional identification and probability-kernel results, reconstructed the Chapter 6 simulation, examined restricted Bayesian diagnostics, and separated numerical correspondence from reconstruction scope. Extracted plot coordinates and calculations using published coefficients reuse source evidence; neither is an independent empirical replication.

The evidence for the simulation is available in `summary.json` and `chapter6-result-mapping.json`. The conditional arguments are recorded in `scale-invariance-evidence.json`, `chapter5-analytical-evidence.json`, `narrow-analytical-evidence.json` and `chapter6-analytical-evidence.json`. These are ordinary mathematical arguments, not Lean-certified theorems.

3 Missing covariates: precision and power

Full information has lower reconstructed variance-component standard errors throughout the primary Chapter 6 grid. We used 1,000 independent datasets in each of 36 primary conditions: three missingness mechanisms, two to five covariates, and missing fractions 0.02, 0.06 and 0.10. Both full-information and

twin-wise deletion methods were fitted to each dataset. Twelve additional conditions at 0.03 missingness are a separate sensitivity analysis. Thus 48,000 datasets produce 96,000 completed method fits, comprising 72,000 primary and 24,000 sensitivity fits. All pass the recorded numerical guards; no fits are excluded. A guard is not a certificate of a global maximum.

The primary precision comparison is deletion minus full-information mean variance-component SE. All 108 differences are positive, and all 108 Bonferroni-adjusted Student intervals exclude zero positively. The illustration (MCAR, two covariates, 2% missingness, A variance) has difference 0.00150234874, paired Monte Carlo standard error (MCSE) 0.00002230935, and adjusted interval [0.00142397666, 0.00158072082]. Its source point difference is 0.00106356032. All 216 literal plotted variance-SE values lie outside the conditional source-comparison bands. The source's SE implementation and Monte Carlo uncertainty were not recovered, so these discrepancies do not establish nonreplication of a fully comparable numerical estimand. The exploratory path-SE difference 0.00106818378 is a different estimand and cannot replace variance SE.

The missingness mechanisms are missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR). Under MAR, deletion bias directions agree with the source in all 36 component-condition cells: A and C are underestimated and residual heritability $h^2 = A/(A+C+E)$ is overestimated. Adjusted intervals exclude zero in 10 of 12 A cells, all 12 C cells, and 9 of 12 heritability cells. A bias spans [-0.07283115,-0.00621756], C bias [-0.11495778,-0.00766858], and heritability bias [0.00141622,0.03118039]. Full-information bias ranges are respectively [-0.00319176,0.00133589], [-0.00261639,0.00101715], and [-0.00139029,0.00265364]; none excludes zero after adjustment. Those null-compatible intervals do not establish zero bias or equivalence.

Eleven full-information point-bias directions reverse relative to the source, but none excludes zero after adjustment. The two conditional MAR source-estimate deviations both occur at the ambiguous low missingness level. The heritability illustration at MAR, five covariates and 10% missingness has bias 0.03118038817, MCSE 0.00304897252, and exploratory adjusted interval [0.02080654405,0.04155423228]; its source bias is 0.03315402096. Full condition-level signed estimates, source deviations and uncertainty are retained in `signed-SE-contrasts.csv` and `signed-bias-contrasts.csv`.

4 Measurement models and Bayesian recovery

The empirical moderation findings cannot be independently refitted from the available inputs. The source mathematics AxE slope is 2.0837 with 95% highest posterior density (HPD) interval [1.0886,3.0344], while AxC is 2.6159 with interval [-0.1358,5.2189] (Schwabe, 2016, Table 3.2). Liberalism AxE is -2.81 [-3.22,-2.39] and AxC is 0.54 [0.31,0.84] (Schwabe, 2016, Table 4.3); the high-SES shared-variance slope is -3.0777 [-7.3886,-0.03] (Schwabe, 2016, Table 5.4). Our corresponding independent estimates are unavailable. Missing NTR/Cito linkage, calibration, responses and joint posterior draws block these comparisons; the Virginia 30K responses needed for Chapter 4 are also unavailable. The source's two-dimensional attitude structure, nine-item subscale, IRT/sum-score reversals, and thirteen-covariate nondetection therefore retain blocked status.

The bounded item-level diagnostic does not establish Bayesian recovery. Four chains used one fixed floor-design dataset with 500 twin pairs, 60 items and generating interaction zero, data seed 301, 1,000 adaptation iterations, 7,000 burn-in iterations and 12,000 retained draws per chain. All chains adapted and completed with finite draws, but six of fourteen monitored columns failed the joint convergence gate: the three variance components and their log transforms. Maximum rank-based $\widehat{R}$ was 1.04647, minimum bulk effective sample size (ESS) 106.11, minimum tail ESS 262.35, and maximum mean MCSE/SD 0.10457. The gate required $\widehat{R} < 1.01$, bulk and tail ESS at least 400, and MCSE/SD at most 0.05 (Vehtari et al., 2021). The interaction mean -0.11216 is a one-dataset diagnostic, not a source 100-dataset

recovery estimate. Source 100-dataset Chapter 2 and 250-dataset Chapter 5 targets, including detection rates and twenty-item sufficiency, are not established. See `jags-gate-scientific-evidence.json`.

5 Conditional analytical results

Common-scale population identification can hold even when item subsets change finite-data likelihood evidence. Under the assumptions below, two distinct conditionally independent calibrated binary observations identify a Gaussian trait's mean and variance; cross-twin joint probabilities identify the interior covariance. With $T = A + C + E$, $K_{MZ} = A + C$ and $K_{DZ} = A/2 + C$, the inverse relations are

$$A = 2(K_{MZ} - K_{DZ}), \qquad C = 2K_{DZ} - K_{MZ}, \qquad E = T - K_{MZ}.$$

Both zygosities at two moderator levels identify the stipulated log-linear variance coefficients. Correct specification and injectivity yield a unique population expected-log-likelihood maximizer by the Kullback–Leibler argument. They do not establish finite-sample likelihood concavity or uniqueness.

An exact finite example shows why this distinction matters. On the same common scale, let P have $(A, C, E) = (1/4, 1/4, 1/2)$ and Q have $(1/2, 1/8, 3/8)$ at both moderator levels. Both have total variance 1 and DZ covariance $3/8$, while MZ covariance rises from $1/2$ to $5/8$. Four distinct Rasch observations per twin have slope 1 and difficulty 0. In MZ families, one two-item subset is all ones in both twins; the other is all ones in one twin and all zeros in the other. Gaussian covariance differentiation (Voigtlaender, 2020, Corollary 2) gives opposite strict likelihood orderings of P and Q for the two subsets. DZ factors cancel. The configuration has positive probability and retains the required support. This is not a claim about opposite global MLEs or significance decisions, and it does not assert that Schwabe intended exact finite-sample invariance.

The logvariance formulation has useful elementary properties without proving likelihood uniqueness. For $V(m) = \exp(b_0 + b_1 m)$, its derivative has the sign of b_1; at zero slope it is constant. For a squared path $(a + bm)^2$ with shared binary moderator, the reflection $b^* = -2a - b$ preserves both endpoints while keeping $a > 0$. Distinct reflected slopes require $a + b \neq 0$. Unrestricted global sign symmetry extends to continuous moderators, but with three distinct support values and a strictly positive intercept, equality of squared affine paths forces equality of the paths. An all-zero response dataset with a free common mean has a likelihood supremum approached as the mean tends to minus infinity, without a finite maximizer. This refutes universal finite-MLE existence, a stronger interpretation than the source wording; uniqueness when a maximum exists remains unresolved (Schwabe, 2016, p.72; Appendix C).

The Chapter 6 Gaussian covariance construction is an exact model identity. If $x = (x_1, x_2)$ has diagonal covariance blocks $S = W + B$ and cross-block B, and $y_j = \mu + \beta^T x_j + u_j$ with residuals independent of x, then $\mathrm{Var}(y_j) = A + C + E + \beta^T S \beta$ and $\mathrm{Cov}(y_1, y_2) = \rho A + C + \beta^T B \beta$. Conditioning on observed variables reduces missing-value conditional variance by a positive semidefinite Schur-complement term. Strict reduction requires nonzero residual conditional covariance. An exact example has successive conditional variances $2, 3/2, 12/11, 279/286$. This known-parameter prediction result does not guarantee smaller estimated biometric SEs or higher rejection-rate power.

6 Methods and comparison rules

Our uncertainty calculations distinguish simulation error from source reconstruction error. MCSE is the across-dataset standard deviation divided by the square root of the number of valid replicates; paired contrasts use within-dataset differences. Two-MCSE bands are descriptive initial comparisons. Bonferroni-adjusted Student intervals control the specified familywise error at 0.05 (Dunn, 1961): 108

primary variance-SE contrasts, 36 sensitivity contrasts, separate exploratory path-SE families of 108 and 36, 216 source-point comparisons per SE metric, and 72 MAR bias comparisons. Shared simulation and parameter dependence is retained; records are not independent trials. Source-comparison bands add the geometric extraction bound to our uncertainty band. They omit unknown source Monte Carlo error and are not full two-sample confidence intervals.

Numerical agreement and reconstruction scope answer different questions. A numerical match rounds half up to the printed source value; "close" differs by at most one unit of its last printed decimal; a larger comparable gap is a mismatch. Scope is exact only for the same input, estimator and procedure, limited for reconstructions, and unavailable for missing inputs. Figure coordinates lack printed decimal precision, so no ordinary rounding verdict is assigned. The pre-production extraction bound uses 0.25 PDF point per coordinate propagated through axis scaling and summed for contrasts. Earlier baseline outcomes were known; the complete plan was recorded after launch but before inspecting the new production effects, so it was not wholly preregistered. The ordering result and conditional bands do not establish source equivalence or direct detection power.

R 4.5.0 supplies statistical execution (R Core Team, 2025); OpenMx 2.21.13 fits the reconstructed Gaussian likelihood (Neale et al., 2016). lavaan 0.6.19 provides an independent baseline covariance fit (Rosseel, 2012). Their baseline variance estimates agree within approximately 5×10^{-8}, while their SEs are not identical. JAGS 4.3.2 with rjags 4-16 is used because the item-level non-Gaussian likelihood is not a covariance-only SEM (Plummer, 2024); CODA supports historical diagnostics (Plummer et al., 2006), and posterior 1.6.1 supplies the rank-based diagnostics. These checks do not independently validate the full moderated posterior. The terminal audit recomputed the same saved trials. The Bayesian outcome audit used the same saved draws and did not independently reimplement ESS or MCSE. Exact algebra checks share the source model assumptions. These are independent checks of specified calculations, not independent end-to-end studies. Exact runtime records are in `runtime-packages.json` and `posterior-runtime-packages.json`.

7 Assumptions

7.1 Reconstructed covariate allocation

Equal covariate coefficients are our reconstruction choice because the original driver and exact allocation are unavailable. The 2% design is primary and 3% is a separate sensitivity case; neither resolves the source's conflicting labels. The Gaussian latent model and observation mechanisms must be correctly specified for their statistical interpretation.

7.2 Common calibrated observation scale

Our population argument assumes known finite positive logistic slopes and finite intercepts on one anchored scalar scale, at least two distinct conditionally independent observations per subset, independent cross-twin measurement errors, and observable within-person and cross-twin joint probabilities. Subsets are prespecified or their selection model is included. Calibration alone does not identify unobserved joint moments.

7.3 Gaussian components and moderator support

The identification bridge assumes strictly positive Gaussian ACE components, common conditional means and variances across zygosity, independent families, genetic correlations 1 and 1/2, and posi-

tive joint support for both zygosities at each of two distinct shared observed moderator levels. Means are linear and component variances log-linear in that moderator. These are sufficient mathematical assumptions, not verified empirical properties. They exclude Chapter 2's non-Gaussian latent-genetic interaction family and broader ordinal or multidimensional measurement models.

7.4 Conditional formulas and numerical checks

Gaussian conditional formulas require nonsingular observed covariance blocks and fixed known parameters. Strict information gain requires nonzero conditional covariance. Numerical guard validity does not establish global optimality, missingness ignorability or population model validity. No formal proof-assistant bridge is supplied.

8 Ambiguities that were resolved

8.1 Variance ratios versus standard-deviation ratios

The source environmental-variance ratio over one genetic standard deviation is $\Delta = \exp(\beta_1\sqrt{V_A})$, hence $\beta_1 = \log(\Delta)/\sqrt{V_A}$ for $V_A > 0$. The SD ratio is $\sqrt{\Delta}$. At $V_A = 0$ this inverse is undefined. The general identity is established, but the Chapter 2 example's change from environmental baseline 0.2 to the multiplication $0.5 \times 1.5 = 0.75$ is not repaired by it. Chapter 3's reported ratios 1.63 and 1.85 correspond to a one-SD shift despite text describing a variance shift. These are source arithmetic/interpretation issues, not independent empirical successes; see narrow-analytical-evidence.json.

8.2 Information on the trait scale

The printed Chapter 3 Eq.3.4 omits the squared discrimination factor. For Bernoulli probability $p = \text{logistic}(a(\theta - b))$, the score is $a(Y - p)$ and information is $a^2p(1 - p)$. At $a = 2, \theta = b$, this is 1 rather than the printed 1/4. The formula is correct for unit slope or the item logit coordinate, not arbitrary common-trait slopes. Information adds under conditional independence and known calibration; inverse square-root information is a conventional information-based SE, not every finite-sample or posterior SD. The plot implementation is unavailable, so a printed discrepancy does not prove an implemented error.

8.3 Ordinal probabilities versus the printed formulas

Appendix B's cumulative-exponential weights normalize and give the stated adjacent-category logits. The literal Chapter 4 formulas instead produce probabilities $1/2, 1/2, 1/4$ at the zero setting, summing to 5/4, while Appendix B gives three probabilities 1/3. Its first cumulative term cancels and effective transitions use the second and third thresholds. This establishes implemented-kernel normalization and a separate printed-formula discrepancy, not empirical reproduction. Evidence and source locators are in narrow-analytical-evidence.json.

8.4 Binary and continuous observation kernels

Appendix G uses probabilities 0.001 and 0.999 around a zero liability threshold, not a deterministic threshold. Marginally $P(X = 1) = 0.001 + 0.998\Phi(\mu/\sigma)$ and independent observation errors multiply binary cross-covariance by 0.998^2. The continuous school-size observation has precision 100, hence residual variance 0.01 on its standardized scale. The phenotypic regression uses observed or imputed X, not

its latent expectation. These semantics differ from the continuous joint-Gaussian simulation and do not establish posterior convergence; see chapter6-analytical-evidence.json.

8.5 Indexing and support of the source designs

Appendix E contains the separate-moderator extension referred to as Appendix F on p.74. The Chapter 6 table lists six pedagogical categories despite prose saying five; its Reformed category label varies. The primary design includes five covariates despite shorter summary lists. Primary Chapter 4 positive AxC values are retained despite the opposite Dutch-summary wording. These are resolved indexing or source-selection decisions, not new experiments. All source conflicts, including unresolved ones, are retained in evidence-packet.json.

9 What remains unresolved

The principal unresolved causes are missing inputs, incomplete validated posterior targets, and source implementation ambiguity. For empirical claims, the public-source search found original model programs and a supplement but no matching responses, linkage or calibration files. Printed values and BayesTwin examples cannot substitute for those inputs. The required settling artifacts are the original datasets, calibration/linkage records, exact model versions and joint posterior draws.

For Bayesian recovery, short and source-length single-dataset diagnostics and a four-chain check were available, but the joint convergence gate failed. Mixing is the immediate demonstrated obstacle; the missing repeated-dataset aggregates remain a separate requirement. The settling evidence is a validated sampling procedure with joint diagnostics followed by the source-count 100/250-dataset designs. A favorable single coefficient or probability cannot settle that family. Historical diagnostic values and their limited scopes are retained in historical-diagnostic-evidence.json.

For Chapter 6 numerical fidelity, the complete fixed grid settles our reconstruction's ordering, but not the original SE calculation. Variance versus path SE and 2% versus 3% are the leading identified ambiguities; an original plotting/fitting driver and its replicate-level SEs would distinguish them. More repetitions of the same reconstruction cannot recover that provenance. The source power assertion requires a rejection-rate experiment, not only smaller SEs.

For the broad scale-invariance and likelihood-uniqueness statements, population identification is conditionally established, but broader measurement and latent-genetic families are not covered. The all-zero example settles only nonexistence under a universal finite-MLE reading. A proof or counterexample under explicit nondegenerate support and parameter constraints is needed for at-most-one finite maximum. Actual empirical calibration and observation assumptions are independently unresolved.

Other printed inconsistencies remain explicit: Chapter 4's heritability mean 0.54 lies outside its printed HPD [0.34,0.51]; Chapter 5 prose says 80,000 pairs while its table/results say 8,000; the low-missingness labels disagree. We do not silently replace these numbers. They are source discrepancies, not demonstrated failures of independent fitted estimates.

9.1 Scope limits of the evidence

L0. Single-dataset posterior runs and short pilots do not reproduce the source 100-dataset Chapter 2 or 250-dataset Chapter 5 summaries. Adequate joint convergence and repeated-dataset uncertainty are not established.

L1. Unavailable original empirical responses, calibration and linkage records, joint posterior draws, and underreported designs prevent the corresponding independent empirical or exact-design comparisons.

L2. Figure geometry and arithmetic from printed parameters reuse source evidence. They do not reproduce the original data generation or empirical fits.

L3. Conditional ordinary mathematical arguments do not supply formal certification or verify empirical calibration, observation and support assumptions.

L4. The precision grid contains 36 primary conditions and 12 additional 3% missingness sensitivity conditions. The source discrepancy between 2% in prose and 3% in plots is unresolved. Source-derived geometric bounds are neither original Monte Carlo uncertainty nor equivalence margins.

L5. The MAR A, C and heritability comparisons contain 11 full-information point-sign reversals; none excludes zero after adjustment. Both conditional MAR source-estimate deviations occur at ambiguous low missingness.

L6. The Chapter 2 IRT bias-remedy and power claims and the Chapter 5 Bayesian claims are not established by the bounded four-chain, one-dataset diagnostic, which fails its joint convergence gate. No source 100- or 250-dataset aggregate, power fraction or empirical effect is inferred from it.

L7. The scale-invariance and Chapter 5/6 analytical results are assumption-bound ordinary arguments, not Lean-certified proofs. Population identification is not finite-sample likelihood uniqueness.

L8. The whole-dissertation inventory contains 996 linked records, including shared components, source-only arithmetic and parents. They are not 996 independent findings, and no replication success fraction is defined.

10 Code and source availability

The dated acquisition ledger below distinguishes accessible source material from unavailable scientific inputs. The citation audit verified software metadata rather than full texts of every software paper; Dunn (1961) was available as publisher metadata and abstract, Vehtari et al. (2021) was checked in Sections 1–2, and the complete primary Price-theorem text was checked. These access limits do not supply missing statistical assumptions. See `citation-audit.json`. The dissertation institutional record verifies its 2016 University of Twente doctoral identity and DOI. The Chapter 4 article and supplement (Schwabe et al., 2016) add plots and code, not the missing responses. BayesTwin version 1.0 (2017) and its pre-defense repository revision were inspected separately; its example posterior is explicitly simulated (Schwabe, 2017). Public searches are bounded searches, not proof that inputs do not exist. Full copyrighted thesis files and unlicensed original programs stay local; they are not redistributed with our replication code.

Dated availability evidence: `scientific-availability.json`.

Date and location	Result
2026-10-30; Institutional final-version record	Verified PhD, institution, advisers, date, DOI, ISBN and final-version designation.
2026-10-30; Author dissertation GitHub repository	Full 164-page thesis and five model programs obtained at `6c2c48901209fc76ba932bb094c61abba5ef160d`. No raw empirical datasets or Monte Carlo driver/seeds. Full text read. Original files kept local; rights reserved/no code license.

Date and location	Result
2026-10-30; All 12 public author GitHub repositories	All cloned; immutable hashes in author-repositories.json. Data-file and code audit found workshop data, BayesTwin simulated posterior example and unrelated projects; no matching NTR/Cito/Virginia study inputs or curvilinear simulation driver. This is a search limit, not proof of absence.
2026-10-30; OSF indexed search	Queries site:osf.io Schwabe twin and exact Inga Schwabe 2016 found later unrelated projects/preprints, no matching dissertation simulation or empirical input. No identified thesis OSF project.
2026-10-30; GitLab indexed search	Queries Schwabe and exact ingaschwabe returned unrelated authors/no matching repository. No claim of exhaustive private-site search.
2026-10-30; Zenodo indexed search	Queries Schwabe twin and exact Inga Schwabe returned unrelated records/later projects; no matching dissertation data/code record.
2026-10-30; Chapter 4 article and supplement	Publisher confirms CC BY 4.0 article and links ten-page supplement. Supplement accessible through browser; contains category plots, fit plots and JAGS program, not raw responses. Original thesis code replayed locally. Supplement findings cannot be refitted without Virginia 30K responses.
2026-10-30; BayesTwin repository history	Current code `656afd72d45ecd13f54b72c9400cceface8d423f`, version 1.0 dated 2017, GPL version 2 or later. Fetched full history; last pre-defense revision `a8e5d6b5ee6a8c76bdd0234a00fbcc809712ac08` dated 2015-10-15 available for separate replay. Example results.rda explicitly simulated, not dissertation empirical data.
2026-10-30; Historical statistical engine specification	Thesis cites R/rjags 2013 and OpenMx 2011/JAGS 2003 publications but no complete sessionInfo, engine patch versions, random seeds or driver. Exact historical environment not determined. Current pinned versions and original-model replay used with explicit scope limit.

11 Scientific compliance ledger

The ledger records substantive requirements and their limitations. A departure denotes an explicit limit of the scientific evidence, not successful whole-dissertation replication. Rows retain the scientific content of `scientific-compliance.json`.

Requirement	Status	Evidence and justification
Whole-dissertation coverage	followed	`coverage-v2.csv` and `evidence-packet.json` cover all chapters and appendices, 996 linked source records and 26 principal claims. Inventory coverage is not independent replication or whole-study success.
Original-source and input audit	followed	`scientific-availability.json` records dated permitted acquisition attempts. All 12 public author repositories were examined. Unavailable empirical inputs and simulation specifications retain unresolved targets.
Preferred statistical engines	followed	`preflight.json` records R 4.5.0, lavaan 0.6.19 and OpenMx 2.21.13. OpenMx implements the Chapter 6 Gaussian likelihood; `engine-agreement.csv` supplies independent baseline checks.

Requirement	Status	Evidence and justification
JAGS for non-Gaussian latent item models	departed	runtime-packages.json; posterior-runtime-packages.json; jags-gate-scientific-evidence.json. Justification: A covariance-only SEM would change the source likelihood. JAGS 4.3.2/rjags 4-16 reconstructs the item-level models; posterior 1.6.1 computes diagnostics. Independent generator, baseline engine and model-semantic checks do not independently validate the full posterior.
Original repetition counts and adequate posterior convergence	departed	summary.json; jags-gate-scientific-evidence.json; resource-projection.json. Justification: Chapter 6 completes 1000 datasets in 36 primary and 12 sensitivity conditions, 96000 method fits. Chapters 2–5 source 100/250 dataset aggregates remain incomplete. The bounded four-chain one-dataset diagnostic completed but failed joint convergence. Short diagnostics and source-only arithmetic are not production estimates.
Signed estimands, uncertainty and multiplicity	followed	analysis-plan.json separates variance/path SE, deletion-minus-full contrasts, estimate-minus-truth bias and source deviations. Geometric extraction margins, our Monte Carlo error and unknown source Monte Carlo error remain distinct; descriptive magnitude is not equivalence.
Analytical verification and formal proof	departed	chapter5-analytical-evidence.json; chapter6-analytical-evidence.json; formalization-environment-check.json. Justification: Ordinary analytical and exact-arithmetic checks are independently reviewed; Lean formalization is absent. Population identification, finite-likelihood uniqueness and empirical inference remain distinct. No toolchain install or source build was attempted; missing local binaries do not prove mathematical or computational impossibility.
Source attribution and permitted redistribution	followed	references.bib and citation-audit.json identify source support. Full copyrighted dissertation and unlicensed original programs stay local; only original replication code, generated results and limited attributed source crops are redistributed.
Reproducible versions and commands	followed	terminal-reproduction.md specifies immutable completed-analysis commit and commands; README.md documents runtime, simulation and diagnostic execution. Navigation links to master are separate from immutable reproduction revisions.

12 Next steps

The most consequential evidence would be the original empirical responses, linkage and calibration files: these would enable direct refits of the blocked moderation and covariate claims. Original replicate-level SEs and plotting code would resolve the Chapter 6 variance/path and low-missingness ambiguities. A sampler with adequate joint convergence would then permit source-count recovery and detection experiments. A separate mathematical argument must settle finite-maximum uniqueness; additional simulation precision cannot prove it.

A Complete coverage inventory

Every one of the 996 linked records maps to this appendix and to its exact identifier in `coverage-v2.csv` and `evidence-packet.json`. The packet retains the source locator, published and reconstructed quantities, source conditions, methods, uncertainty, dependencies, numerical and reconstruction decisions, limitations, and stopping evidence for every claim. This is a complete inventory mapping, not an assertion that each record is replicated. The 26 principal claims appear on page 1; the 187 component comparisons are evidence-only and retain all parent contrast indices and their own tolerances and uncertainties. They are not additional independent findings.

B Principal comparison details

The following table separates scientific findings, numerical precision and reconstruction scope. "Withheld" denotes an unavailable or incomparable numerical target; it is not numerical mismatch. The source locators refer to Schwabe (2016). A proposition has no numeric rounding target.

Claim / source	Published	Ours	Precision / numerical	Scope
HE01; Chapter 1 p 5; Chapter 4 p 51; Chapter 7 pp 107–108	Under a common appropriately identified IRT measurement scale, biometric conclusions are asserted to be independent of the particular item subset.	Admissible subsets identify the same population parameter under the stated calibrated Gaussian ACE assumptions. Two admissible subsets can give opposite finite likelihood orderings of the same two ACE parameter points.	No scalar target	limited
HE02; Chapter 2 abstract p 11; aim p 24; discussion p 25.	Incorporating an IRT measurement model remedies the scale-induced interaction bias and recovers generating parameters in the studied designs.	Independent target not established.	No scalar target	limited
HE03; Chapter 2 abstract p 11; results p 22; discussion p 25.	The proposed method has useful power to detect genotype–environment interaction under the investigated sample-size and effect-size conditions.	Independent target not established.	No scalar target	limited
HE04; Chapter 2 abstract p 11; results pp 21–22; discussion pp 24–25.	Sum-score analysis can produce spurious genotype–environment interaction from measurement-scale properties.	Independent target not established.	No scalar target	limited

Claim / source	Published	Ours	Precision / numerical	Scope
HE05; Chapter 3 p 42, Table 3.2	2.0837	Independent target not established.	Withheld; half unit 0.00005	unavailable
HE06; Chapter 3 p 42, Table 3.2	2.6159	Independent target not established.	Withheld; half unit 0.00005	unavailable
HE07; Chapter 4 abstract p 47, pp 58–61, Figs 4.1–4.4	The 28-item scale measures two distinct dimensions: liberalism–conservatism and use of the question-mark response.	Independent target not established.	No scalar target	unavailable
HE08; Chapter 4 pp 62, 66	Nine selected items constitute the proposed ordinal social-liberalism subscale.	Independent target not established.	No scalar target	unavailable
HE09; Chapter 4 p 63, Table 4.3	-2.81	Independent target not established.	Withheld; half unit 0.005	unavailable
HE10; Chapter 4 p 63, Table 4.3	0.54	Independent target not established.	Withheld; half unit 0.005	unavailable
HE11; Chapter 4 pp 63–64, 66–67, Tables 4.3–4.4	The interaction point-estimate direction changes between IRT and sum-score analyses: β_1	Independent target not established.	No scalar target	unavailable
HE12; Chapter 4 pp 63–64, 66–67, Tables 4.3–4.4	The interaction point-estimate direction changes between IRT and sum-score analyses: gamma 1	Independent target not established.	No scalar target	unavailable
HE13; Chapter 5 pp 69, 72–77, 83–84; Appendix C pp 138–139	The alternative logvariance ACE×M model is asserted to be uniquely identified.	Reviewed ordinary joint-item argument establishes population injectivity under known finite positive item slopes, distinct locally independent observations, positive-interior Gaussian ACE and both zygosities at each of 2 moderator levels.	No scalar target	limited
HE14; Chapter 5 pp 69, 72–77, 83–84; Appendix C pp 138–139	Simultaneous IRT and ACE×M modelling is asserted to prevent spurious moderation arising from heterogeneous measurement error.	Independent target not established.	No scalar target	limited

Claim / source	Published	Ours	Precision / numerical	Scope
HE15; Chapter 5 pp 69, 72–77, 83–84; Appendix C pp 138–139	The source states the logvariance model has only one maximum in its likelihood function.	Independent target not established.	No scalar target	limited
HE16; Chapter 5 abstract p 69, pp 78, 83	Large samples are said to be necessary to detect β_{1a} moderation.	Independent target not established.	No scalar target	limited
HE17; Chapter 5 abstract p 69, pp 78, 83	Large samples are said to be necessary to detect β_{1c} moderation.	Independent target not established.	No scalar target	limited
HE18; Chapter 5 pp 80, 83	As few as 20 phenotypic items are asserted sufficient to find moderation effects.	Independent target not established.	No scalar target	limited
HE19; Chapter 5 p 82, Table 5.4	-3.0777	Independent target not established.	Withheld; half unit 0.00005	unavailable
HE20; Chapter 6 abstract p 86; results p 99 and Table 6.3 p 100; discussion p 102.	None of the thirteen measured covariate coefficients was detected in the educational-achievement application using the reported 99.6% HPD criterion.	Independent target not established.	No scalar target	unavailable
HE21; Chapter 6 abstract pp 85–86; pp 91, 101–102; Chapter 7 p 104.	The source asserts that full information is more powerful than deletion regardless of the investigated missingness mechanism, covariate count or amount of missingness.	Independent target not established.	No scalar target	limited
HE22; Chapter 6 results pp 93–95; discussion p 101.	Full information gives lower variance-component standard errors than deletion across the investigated missingness mechanisms and design grid.	All 108/108 signed deletion-minus-full variance-SE differences are positive; all 108/108 Bonferroni-adjusted intervals exclude 0 positively. The qualitative precision ordering is reproduced for this complete component grid.	No scalar target	limited

Claim / source	Published	Ours	Precision / numerical	Scope
HE23; Chapter 6 pp 88–91; Appendix F	The full-information model uses all observed phenotype and covariate components when covariate data are incomplete.	Independent linear-transformation covariance construction agrees with Eq 6.5 for q 2–5; the OpenMx raw-data normal likelihood implements joint observed-data fitting with missing values, and full-information/deletion arms both completed the frozen grid.	No scalar target	limited
HE24; Chapter 6 pp 94–95, Fig 6.5	MAR deletion underestimates A variance while full information remains close to the generating value.	Deletion bias is negative in 12/12 conditions, with 10/12 adjusted intervals excluding 0 in that direction; bias range [-0.07283115, -0.00621756]. Full-information bias range [-0.00319176, 0.00133589], with 0/12 adjusted intervals excluding 0.	No scalar target	limited
HE25; Chapter 6 pp 94–95, Fig 6.5	MAR deletion underestimates C variance while full information remains close to the generating value.	Deletion bias is negative in 12/12 conditions, with 12/12 adjusted intervals excluding 0 in that direction; bias range [-0.11495778, -0.00766858]. Full-information bias range [-0.00261639, 0.00101715], with 0/12 adjusted intervals excluding 0.	No scalar target	limited
HE26; Chapter 6 pp 94–95, Fig 6.5	MAR deletion overestimates heritability while full information remains close to the generating value.	Deletion bias is positive in 12/12 conditions, with 9/12 adjusted intervals excluding 0 in that direction; bias range [0.00141622, 0.03118039]. Full-information bias range [-0.00139029, 0.00265364], with 0/12 adjusted intervals excluding 0.	No scalar target	limited

C Artifacts and reproducible commands

The completed Chapter 6 analysis is reproducible from saved reports at the separately labeled immutable Reproduction revision (immutable snapshot). File links below are master navigation paths; they do not assert that branch additions are currently published there. The evidence packet is frozen at commit d1b61b8b1d7af6a17341a2b5fee73a134a35c47c, SHA256 `41c12d8a490b572f1e5ea66a2e12e0becfad1398962616fdbad79b8f4ab2e554`.

From `formalization/schwabe-psychometric-genetics` at the reproduction revision, with Python, NumPy and SciPy installed, regenerate analysis and report tables into fresh directories:

```
python3 simulation/analyze_production.py \
 --reports production/chapter6-v1/reports \
 --output-dir /tmp/schwabe-analysis-replay \
 --source outputs/source-figure-values.csv \
 --plan production/analysis-plan.json
python3 simulation/build_production_report.py \
 --analysis-dir /tmp/schwabe-analysis-replay \
 --output-dir /tmp/schwabe-report-replay
```

These commands analyze saved aggregates; they do not run model fits or re-extract the dissertation. Numerical CSVs should agree with the saved terminal analysis; timestamps and output paths are not numerical targets. A rerun of model fits additionally requires the pinned R/OpenMx runtime, protocol, runner and seed manifest. Original empirical refits cannot be reproduced without their missing inputs. The exact saved-analysis recipe is `terminal-reproduction.md`.

Claim or artifact	File and reproduction method
All 996 source records	`coverage-v2.csv` Frozen inventory; inspect identifiers, not generated model fits.
Scientific comparisons	`evidence-packet.json` Frozen evidence, not new simulation output.
Chapter 6 analysis	`analyze_production.py` Run the exact saved-report command above.
Chapter 6 report	`build_production_report.py` Run the exact table-generation command above.
Simulation protocol	`protocol.json` Model-fit configuration; distinct from saved-analysis replay.
Uncertainty families	`analysis-plan.json` Input to analyze_production.py.
Source figure coordinates	`source-figure-values.csv` Source extraction input; original local source is required to re-extract.
Chapter 6 algebra	`chapter6_analytical_check.py` python3 simulation/chapter6_analytical_check.py; standard library; writes outputs/chapter6-analytical-checks.json.
Narrow algebra	`analytic_checks.py` python3 simulation/analytic_checks.py; inspect pinned dependencies and outputs/analytic-checks.json.
Bayesian diagnostic	`jags-gate-scientific-evidence.json` Frozen four-chain evidence; not a source-count aggregate.

Claim or artifact	File and reproduction method
Editable manuscript	standalone.tex Compile with the command below.
Bibliography	references.bib BibTeX source used in the build below.

With TeX Live, pdfLaTeX and BibTeX, build this source from the study root:

```
cd paper
pdflatex -interaction=nonstopmode -halt-on-error \
 -jobname=whole_dissertation_replication__schwabe_psychometric_genetics \
 standalone.tex
bibtex whole_dissertation_replication__schwabe_psychometric_genetics
pdflatex -interaction=nonstopmode -halt-on-error \
 -jobname=whole_dissertation_replication__schwabe_psychometric_genetics \
 standalone.tex
pdflatex -interaction=nonstopmode -halt-on-error \
 -jobname=whole_dissertation_replication__schwabe_psychometric_genetics \
 standalone.tex
```

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